

1.6.1 Functional Dependency

Let R be the relational schema and x, y be the non-empty set of attributes. Consider t_1, t_2 are any tuples of relation then $x \rightarrow y$ (y functionally determined by x):

If $\forall t_1, t_2 \ t_1 \cdot x = t_2 \cdot x$ then $t_1 \cdot y = t_2 \cdot y$

1.6.2 Types of Functional Dependency

1. Trivial Functional Dependency

Consider relational schema $R(ABCD)$

A FD $x \rightarrow y$ is trivial FD only if $x \supseteq y$.

Example :

$AB \rightarrow A$

$A \rightarrow A$

$B \rightarrow B$

2. Non-trivial Functional Dependency

Consider relational schema $R(ABCD)$

A FD $x \rightarrow y$ is non-trivial only if

$x \cap y = \emptyset$ means no common attributes between x and y attribute sets.

Example :

$A \rightarrow B$

$AB \rightarrow CD$

3. Semi-Non-Trivial Functional Dependency

A combination of both trivial FD and non-trivial FD.

Example :

$A \rightarrow AB \equiv \{A \rightarrow A, A \rightarrow B\}$

1.6.3 Attribute Closure (X^+)

The set of all possible attributes determined by x .

Example :

$R(ABCD) \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$

$\therefore (A)^+ = \{A, B, C, D\}$

1.6.4 Armstrong's Axioms

1. **Reflexive** : If $x \supseteq y$ then $x \rightarrow y$ or $x \rightarrow x$
2. **Transitivity** : If $x \rightarrow y$ and $y \rightarrow z$ then $x \rightarrow z$
3. **Augmentation** : If $x \rightarrow y$ then $xz \rightarrow yz$
4. **Splitting** : If $x \rightarrow yz$ then $x \rightarrow y, x \rightarrow z$
5. **Union** : If $x \rightarrow y$ and $x \rightarrow z$ then $x \rightarrow yz$
6. **Pseudo transitivity** : If $x \rightarrow y, yw \rightarrow z$ then $xw \rightarrow z$

1.6.5 Finding Candidate Key [Minimal Super Key]

X is candidate key of R

If and only if

1. x must be super key of relation R
 $(x)^+ = \{\text{determine all attributes of R}\}$
2. No proper subset of 'x' is super key of relation R.
 $\forall y \subset x : (y)^+ = \{\text{not determine all attributes of R}\}$
- $x \rightarrow y$ is a non-trivial FD in R with y is a prime attribute then relation R has atleast two candidate key.

$(x \rightarrow y)$: Non – trivial FD

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Prime-attribute

1.7 Membership of FD

A FD $x \rightarrow y$ is member (logically implied) of FD set F if and only if $(x)^+$ should determine 'y' in FD set F.

Example : Given FD Set F : { }

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$x \rightarrow y$ FD belongs to F or not ?

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$x^+ = \{\dots y \dots\}$

then $x \rightarrow y$ is member of F.

1.7.1 Testing of Two FD Sets whether they are equal or not

F and G FD sets equality test:

F and G FD sets logically equal if and only

1. **F cover's G** : All FD's of G set must be member's of F set.

$F \supseteq G$

2. **G cover's F** : All FD's of F set must be member's of G set.

$F \subseteq G$

F Cover G	G Cover F	Result
True	True	$F \equiv G$
True	False	$F \supset G$
False	True	$F \subset G$
False	False	F & G are not comparable

1.7.2 Minimal Cover or Canonical Cover

- Minimal Cover of given FD Set (F) is minimum possible FD's (F_m), which is logically equal to 'F' : ($F = F_m$)
- Remove extraneous attribute (useless attribute) from each determinant of FD set F.

$wxy \rightarrow z$

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Extraneous
attribute

$\{wxy \rightarrow z, w \rightarrow x\} \equiv \{wy \rightarrow z, w \rightarrow x\}$

- Every FD must be simple (RHS of any FD should have single attribute).